

Problem: Procedural Checkout System

Write a Java program that calculates the total cost of a customer's purchase.

A purchase contains:

- a product price
- a quantity
- information about whether the customer is a premium customer
- information about whether the customer has a coupon

The checkout system should calculate the total using the following rules:

1. Calculate the subtotal:

`subtotal = price × quantity`

2. Premium customers receive a 10% discount on the subtotal.
3. Customers with a coupon receive an additional \$5 discount.
4. After discounts are applied, calculate an 8% sales tax.
5. Return or display the final total.

For example, a customer purchases 3 items that cost \$20 each.

The customer is a premium customer and also has a coupon.

Your program should calculate the subtotal, discounts, tax, and final total.

For this version, solve the problem procedurally.

You may use variables, conditionals, and static methods. You should not create classes such as `Product`, `Customer`, `Order`, or `DiscountPolicy`.

After the program is working, consider how the design would change if the checkout system later needed to support:

- multiple products
- regular, premium, and employee customers
- different discount rules

- different tax rules
- coupons with different values
- seasonal promotions

Think about which parts of the program would become difficult to maintain as these requirements grow.